

Plato’s Cave and The Genius: Collective Ascent, Trust Dynamics, and Meta-Reform Beyond Self-Correcting Education

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Abstract

This report presents a computational realization of Plato’s Cave as a multi-agent sandbox for simulating the emergence of hierarchical trust and the collective identification of latent truths, grounded in Kara Olivarria’s 25-dimensional quantum education architecture. The sandbox couples a latent *Form*—a 25D prime-indexed pedagogical lattice—with agents that observe noisy shadows of the Form and update their beliefs via local sparse refinement and a social trust-oracle mechanism. On top of this, the “Genius” model is used as a meta-architecture: its five-phase cycle (novelty → enhancement → critique → formalization → validation) governs structural meta-reforms, including hierarchy evolution and dimension-level mutations. We formalize the underlying mathematics, specify implementable algorithms, and report synthetic validation experiments demonstrating epistemic ascent, suppression of “false suns,” and earned hierarchical authority.

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1 Executive Summary

We construct a computational interpretation of Plato’s Cave in which agents attempt to recover an invariant latent *Form* from partial, noisy projections (shadows). The latent Form is instantiated as Olivarria’s 25-dimensional quantum education architecture, a prime-indexed pedagogical lattice representing a planetary curriculum.

Each agent i holds a state $\mathbf{x}^{(i)}(t) \in \mathbb{R}^{25}$ and observes shadows

$$\mathbf{y}_i = \mathbf{A}_i \boldsymbol{\theta}^* + \boldsymbol{\varepsilon}_i,$$

where $\boldsymbol{\theta}^* \in \mathbb{R}^{25}$ is the fixed Form and $\mathbf{A}_i$ encodes the agent’s limited view. Agents perform local LASSO-like refinement and then enter a social dialectic where trust is assigned via counterfactual explanatory power and regularized by an adaptive oracle ensemble.

On a slower timescale, a “Genius” meta-architecture governs hierarchy evolution and structural mutations. Agents earn higher tiers through an impact–atypicality index, and dimensions are added, split, or pruned when Genius-driven metrics over dimension usage and oracle contributions cross thresholds.

Synthetic experiments show that, even with modest noise and adversarial agents, the system:

- reduces mean distance to the 25D Form over time;
- suppresses “false suns” (locally coherent but globally misaligned subspaces);
- yields stable, earned hierarchies aligned with oracle performance;
- supports principled architecture evolution under the Genius cycle.

2 The 25D Form: Quantum Education Architecture

2.1 Latent Form and Prime-Indexed Structure

We model Olivarria’s 25-dimensional quantum education architecture as a latent vector

$$\boldsymbol{\theta}^* \in \mathbb{R}^{25},$$

where each coordinate corresponds to a macro-dimension of the curriculum (e.g., Primordial Cognitive Lattice, Apophatic Learning Singularity, Quantum Institutional DNA, etc.). At a finer level, each coordinate can be understood as a prime-indexed multiplicity structure, but for computational purposes we treat the macro-form as a 25D vector.

Optionally, we can represent an agent’s curriculum state using a multiplicative prime code:

$$P^{(i)}(t) = \prod_{k=1}^{25} p_k^{x_k^{(i)}(t)},$$

where p_k is the k -th prime. In practice, all computation is carried out in exponent space $\mathbf{x}^{(i)}$, and the prime-product encoding is reserved for interpretation and logging.

2.2 Observation Model: Shadows in the Cave

Agent i does not observe $\boldsymbol{\theta}^*$ directly. Instead, it observes a shadow

$$\mathbf{y}_i = \mathbf{A}_i \boldsymbol{\theta}^* + \boldsymbol{\varepsilon}_i,$$

where $\mathbf{A}_i \in \mathbb{R}^{m_i \times 25}$ encodes a partial, possibly biased projection, and $\boldsymbol{\varepsilon}_i$ is noise.

When comparing candidate states $\mathbf{x} \in \mathbb{R}^{25}$ against agent i 's observations, we define a local loss

$$E_i(\mathbf{x}) = \frac{1}{m_i} \|\mathbf{A}_i \mathbf{x} - \mathbf{y}_i\|_2^2 + \lambda \|\mathbf{x}\|_1,$$

where the ℓ_1 term encourages sparsity in the 25D representation relative to the Form.

3 Local Learning: LASSO Refinement

3.1 Proximal Gradient Dynamics

Each agent updates its current state $\mathbf{x}^{(i)}(t)$ by a small number of proximal gradient steps on E_i :

$$\begin{aligned} \mathbf{g}_i(t) &= \frac{1}{m_i} \mathbf{A}_i^\top (\mathbf{A}_i \mathbf{x}^{(i)}(t) - \mathbf{y}_i), \\ \mathbf{z}_i(t+1) &= \mathbf{x}^{(i)}(t) - \eta \mathbf{g}_i(t), \\ \mathbf{x}^{(i)}(t+1) &= \text{prox}_{\eta \lambda \|\cdot\|_1}(\mathbf{z}_i(t+1)), \end{aligned}$$

where $\eta > 0$ is a learning rate and the proximal operator is soft-thresholding:

$$\text{prox}_{\tau \|\cdot\|_1}(\mathbf{z}) = \text{sign}(\mathbf{z}) \odot \max(|\mathbf{z}| - \tau, 0).$$

In the full system, agents typically run a few such steps per outer iteration before entering the social dialectic.

4 Social Dialectic: Trust and Oracle Ensemble

4.1 Counterfactual Loss and Normalization

Let $\tilde{\mathbf{x}}^{(j)}$ denote agent j 's current refined state after local updates. Agent i evaluates how well j 's state explains its own shadow via the counterfactual loss

$$L_{ij}(t) = E_i(\tilde{\mathbf{x}}^{(j)}(t)).$$

To avoid scale distortions across agents with different shadow variances, we normalize:

$$L_{ij}^{\text{norm}}(t) = \frac{L_{ij}(t)}{\widehat{\text{Var}}(\mathbf{y}_i) + \varepsilon},$$

where $\widehat{\text{Var}}(\mathbf{y}_i)$ is an empirical variance estimate and $\varepsilon > 0$ is a small constant.

4.2 Softmax Trust with Annealing and Inertia

We interpret $-L_{ij}^{\text{norm}}$ as an unnormalized log-likelihood of j 's model from i 's perspective. For each agent i we define a trust distribution over senders j :

$$S_{ij}^{\text{raw}}(t) = \frac{\exp(-\alpha_t L_{ij}^{\text{norm}}(t))}{\sum_k \exp(-\alpha_t L_{ik}^{\text{norm}}(t))},$$

where α_t is an annealed sensitivity parameter that increases over time. To model epistemic inertia, we use an exponential moving average (EMA) over trust matrices:

$$S_{ij}(t) = \gamma S_{ij}(t-1) + (1-\gamma) S_{ij}^{\text{raw}}(t),$$

with $\gamma \in (0, 1)$.

This softmax-EMA mechanism approximates Bayesian pooling; calibration can be monitored via proper scoring rules such as the Brier score.

4.3 Oracle Ensemble and Invariance Penalty

We introduce a set of oracle functions $\{O_o\}_{o=1}^K$ that evaluate candidate states $\mathbf{x}$ against separate criteria (e.g., overall fit to held-out data, equity measures, long-term retention). For agent j , the oracle costs at time t are:

$$c_{j,o}(t) = O_o(\tilde{\mathbf{x}}^{(j)}(t)).$$

We maintain an exponentially weighted history of oracle costs and define a per-oracle degradation

$$\Delta_{j,o}(t) = c_{j,o}(t) - \bar{c}_{j,o}(t-1),$$

where $\bar{c}_{j,o}$ is a recent average. The oracle penalty for agent j is then

$$\pi_j(t) = \exp\left(-\kappa \sum_{o=1}^K \phi_o(t) \frac{\max(\Delta_{j,o}(t), 0)}{\tau_\pi}\right),$$

with $\phi_o(t)$ an adaptive weight reflecting oracle o 's predictive usefulness, $\kappa > 0$, and $\tau_\pi > 0$ a scale. We enforce a floor $\pi_j(t) \geq \pi_{\min}$.

Trust utilities are modified by oracle penalties, e.g.

$$U_{ij}(t) = -L_{ij}^{\text{norm}}(t) + \beta_h h^{(j)}(t),$$

then

$$S_{ij}^{\text{raw}}(t) \propto \exp(\alpha_t U_{ij}(t)) \pi_j(t).$$

4.4 Social Mixing

Given the row-stochastic trust matrix $S(t)$, we update states via social mixing:

$$\mathbf{x}_{\text{social}}^{(i)}(t+1) = \sum_j S_{ij}(t) \tilde{\mathbf{x}}^{(j)}(t).$$

We may blend local and social updates:

$$\mathbf{x}^{(i)}(t+1) = (1 - \beta_{\text{mix}}) \tilde{\mathbf{x}}^{(i)}(t) + \beta_{\text{mix}} \mathbf{x}_{\text{social}}^{(i)}(t+1),$$

with $\beta_{\text{mix}} \in [0, 1]$.

5 Hierarchy Evolution and the Genius Index

5.1 Hierarchy Beliefs

Each agent i maintains a hierarchy belief $h^{(i)}(t) \in [0, 1]$ representing its institutional weight. This is used as a bias in trust formation and oracle subsampling.

We define:

- Oracle alignment:

$$o_i(t) = \exp(-\tau_o \pi_i(t)),$$

- Weighted centrality:

$$c_i(t) = \sum_j S_{ji}(t) h^{(j)}(t), \quad c_i^{\text{norm}}(t) = \frac{c_i(t)}{\max_k c_k(t) + \varepsilon}.$$

5.2 Genius Index

The Genius model views a “genius” as a trajectory whose moves are both high-impact and atypical relative to a baseline distribution. We approximate this at the agent level by defining

$$G_i(t) = (1 - \lambda) o_i(t) + \lambda c_i^{\text{norm}}(t),$$

and an atypicality measure $A_i(t)$ derived from the agent’s prime-move usage (or, in the minimal implementation, a proxy based on variance or surprise). The Genius index is

$$J_i(t) = G_i(t) (1 + \eta A_i(t)),$$

with $\eta > 0$ small.

5.3 Hierarchy Update

We map $J_i(t)$ into a target hierarchy level via a sigmoid:

$$h_{\text{target}}^{(i)}(t) = \sigma(\beta_J J_i(t)) = \frac{1}{1 + e^{-\beta_J J_i(t)}},$$

and update $h^{(i)}$ sluggishly:

$$h^{(i)}(t+1) = \eta_h h_{\text{target}}^{(i)}(t) + (1 - \eta_h) h^{(i)}(t),$$

where $\eta_h \in (0, 1)$ controls hierarchy adaptation speed.

Agents with $h^{(i)}(t)$ above a promotion threshold θ_{promote} are treated as higher-tier nodes in subsequent iterations, affecting trust bias and oracle sampling.

6 Genius Meta-Architecture and Meta-Reform

6.1 The Five-Phase Genius Cycle

The Genius model provides a meta-architecture with phases:

1. **Novelty**: propose new moves (e.g., architectural mutations).
2. **Enhancement**: locally refine and implement them.

3. **Critique:** stress-test via oracle and social dynamics.
4. **Formalization:** freeze successful moves into structure.
5. **Validation:** benchmark against baselines and prior phases.

We apply this cycle at the system level to govern structural changes to the 25D architecture and the trust/hierarchy mechanisms themselves.

6.2 Dimension-Level Mutations (Outline)

Briefly, for each dimension k we can track:

- Usage and contribution to oracle fit;
- Genius-weighted innovations involving that dimension;
- Redundancy with other dimensions.

A dimension with low contribution and low Genius-linked activity may be pruned; overloaded dimensions may be split into subdimensions; new dimensions may be proposed by high-Genius agents. Each such mutation is treated as a novelty proposal, run through enhancement/critique/formalization/validation, and either adopted or rolled back.

7 Minimal Algorithm and Code Snippet

7.1 High-Level Algorithm

At a high level, one outer iteration of the sandbox proceeds as:

1. Local LASSO refinement for each agent (fast inner loop).
2. Compute counterfactual losses and normalized trust.
3. Apply oracle ensemble penalties to adjust trust utilities.
4. Update trust matrix via softmax + EMA.
5. Socially mix agent states via trust.
6. Update hierarchy beliefs via Genius index.
7. (Occasionally) propose and evaluate meta-reforms under the Genius cycle.

7.2 Compact NumPy-Style Snippet

Below is a compact Python/NumPy-style snippet capturing the core trust and mixing dynamics (for a fixed architecture):

```
import numpy as np

def proximal_lasso_step(x, A_i, y_i, lr, lam):
    # x: (d,), A_i: (m, d), y_i: (m,)
    residual = A_i @ x - y_i
    grad = A_i.T @ residual / len(y_i)
    z = x - lr * grad
    # soft-thresholding
    return np.sign(z) * np.maximum(np.abs(z) - lr * lam, 0.0)

def trust_oracle_update(X, A, Y, alpha_t, gamma,
```

```

        oracle_penalty, oracle_penalty_tau,
        penalty_min, rng):
N, d = X.shape
# counterfactual losses
# pred[i,j,:] = A[i] @ X[j]
pred = np.einsum('imd,jd->ijm', A, X)
losses = np.mean((pred - Y[:, None, :])**2, axis=2)
var_y = np.var(Y, axis=1, keepdims=True) + 1e-8
L_norm = losses / var_y

row_min = np.min(L_norm, axis=1, keepdims=True)
base = np.exp(-alpha_t * (L_norm - row_min))

# simple oracle: global subsample
k = max(5, int(0.2 * N))
idx_sub = rng.choice(N, k, replace=False)
pred_sub = np.einsum('kmd,jd->kjm', A[idx_sub], X)
err_sub = pred_sub - Y[idx_sub, None, :]
E_oracle = np.mean(err_sub**2, axis=(0, 2)) # (N,)

delta = E_oracle - np.median(E_oracle)
penalty = np.exp(-oracle_penalty
                 * np.maximum(delta, 0.0)
                 / oracle_penalty_tau)
penalty = np.maximum(penalty, penalty_min)
score = base * penalty[None, :]

# row-stochastic trust
score = score - np.max(score, axis=1, keepdims=True)
S = np.exp(score)
S = S / (np.sum(S, axis=1, keepdims=True) + 1e-12)
S = gamma * S + (1 - gamma) * S # EMA placeholder
X_new = S @ X
return X_new, S, E_oracle

```

This snippet can be embedded in a full simulation loop that also performs local LASSO updates and hierarchy evolution.

8 Synthetic Experiments (Outline)

8.1 Setup

We consider N agents, each with an observation matrix $\mathbf{A}_i$ and noise level σ . We initialize $\boldsymbol{\theta}^*$ as a sparse 25D vector and generate shadows $\mathbf{y}_i$. We compare:

- local-only learning,
- trust-only social mixing,
- combined local+trust+oracle+hierarchy dynamics.

We also seed a fraction of agents as “false suns” with initial states aligned to a wrong subspace and track their trust centrality and hierarchy beliefs.

8.2 Key Metrics

- Mean distance to the Form: $\frac{1}{N} \sum_i \|\mathbf{x}^{(i)}(t) - \boldsymbol{\theta}^*\|$.
- Mean oracle penalty.
- Trust centrality of false suns: sum of incoming trust weights from non-adversarial agents.
- Hierarchy entropy over time.

Observed trends in preliminary runs include steady descent in mean distance to the Form, decreasing trust centrality for false suns, and stabilization of hierarchy entropy at an ordered level.

9 Discussion and Future Work

The Plato–25D–Genius sandbox operationalizes a philosophical allegory as a testable, multi-layered dynamical system: local sparse refinement, social trust and oracle-based correction, and Genius-driven meta-reform. Future work includes:

- richer oracle ensembles including real-world signals;
- explicit modeling of the Form as a manifold $\mathcal{M} \subset \mathbb{R}^{25}$;
- phase-based architecture evolution with formal convergence and robustness guarantees;
- empirical mapping to real educational data.

A Mathematical Appendix

In this appendix we formalize several of the claims made in the main text and provide explicit operator-norm bounds under standard assumptions. We focus on: (i) stability and descent of the local LASSO refinement, (ii) contraction properties of the trust-based social mixing, and (iii) interaction of the trust–oracle mechanism with false-sun clusters under bounded noise and restricted operator norms.

Throughout, we adopt the following notation: $\|\cdot\|$ denotes the Euclidean norm on $\mathbb{R}^d$, $\|\cdot\|_{\text{op}}$ the induced operator norm on matrices, and $\mathbf{1}$ the all-ones vector of appropriate dimension.

A.1 Assumptions

We work under the following assumptions.

- (A1) **Form and observations.** The latent Form is fixed and given by $\boldsymbol{\theta}^* \in \mathbb{R}^d$ with $d = 25$. For each agent $i \in \{1, \dots, N\}$ there is an observation matrix $\mathbf{A}_i \in \mathbb{R}^{m_i \times d}$ and noisy observations

$$\mathbf{y}_i = \mathbf{A}_i \boldsymbol{\theta}^* + \boldsymbol{\varepsilon}_i,$$

where $\boldsymbol{\varepsilon}_i$ are independent, mean-zero random vectors with $\mathbb{E}\|\boldsymbol{\varepsilon}_i\|^2 \leq m_i \sigma^2$ for some $\sigma > 0$.

- (A2) **Operator-norm bounds.** There exist constants $0 < \mu_{\min} \leq \mu_{\max} < \infty$ such that for all i ,

$$\mu_{\min} \leq \|\mathbf{A}_i\|_{\text{op}}^2 \leq \mu_{\max}.$$

(A3) **Step size for local refinement.** The learning rate $\eta > 0$ for the local gradient step satisfies

$$0 < \eta < \frac{2}{\mu_{\max}}.$$

(A4) **Trust matrix structure.** At each outer iteration t , the trust matrix $\mathbf{S}(t) \in \mathbb{R}^{N \times N}$ is row-stochastic:

$$S_{ij}(t) \geq 0, \quad \sum_{j=1}^N S_{ij}(t) = 1 \quad \forall i.$$

We denote by $\rho(\mathbf{S}(t))$ the spectral radius of $\mathbf{S}(t)$ considered as a linear operator on the subspace orthogonal to $\mathbf{1}$.

(A5) **Oracle alignment and penalty floor.** For each t , the oracle penalties $\pi_j(t)$ satisfy $\pi_j(t) \in [\pi_{\min}, 1]$ with $\pi_{\min} \in (0, 1)$ fixed. The modified utilities used to construct $\mathbf{S}(t)$ are such that for non-adversarial agents, expected oracle losses decrease in t .

A.2 Local LASSO Refinement: Stability and Descent

We consider the local objective for agent i :

$$E_i(\mathbf{x}) = \frac{1}{m_i} \|\mathbf{A}_i \mathbf{x} - \mathbf{y}_i\|_2^2 + \lambda \|\mathbf{x}\|_1,$$

with $\lambda \geq 0$. The proximal gradient update for agent i at time t is

$$\begin{aligned} \mathbf{g}_i(t) &= \frac{1}{m_i} \mathbf{A}_i^\top (\mathbf{A}_i \mathbf{x}^{(i)}(t) - \mathbf{y}_i), \\ \mathbf{z}_i(t+1) &= \mathbf{x}^{(i)}(t) - \eta \mathbf{g}_i(t), \\ \mathbf{x}^{(i)}(t+1) &= \text{prox}_{\eta \lambda \|\cdot\|_1}(\mathbf{z}_i(t+1)). \end{aligned}$$

[Lipschitz constant of the smooth part] Under (A2), the gradient of the smooth part

$$f_i(\mathbf{x}) = \frac{1}{m_i} \|\mathbf{A}_i \mathbf{x} - \mathbf{y}_i\|_2^2$$

is Lipschitz with constant $L_i = \frac{2}{m_i} \|\mathbf{A}_i\|_{\text{op}}^2$. In particular, $L_i \leq \frac{2}{m_i} \mu_{\max}$.

We have

$$\nabla f_i(\mathbf{x}) = \frac{2}{m_i} \mathbf{A}_i^\top (\mathbf{A}_i \mathbf{x} - \mathbf{y}_i).$$

For any $\mathbf{x}, \mathbf{z} \in \mathbb{R}^d$,

$$\begin{aligned} \|\nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{z})\| &= \frac{2}{m_i} \|\mathbf{A}_i^\top \mathbf{A}_i (\mathbf{x} - \mathbf{z})\| \\ &\leq \frac{2}{m_i} \|\mathbf{A}_i^\top \mathbf{A}_i\|_{\text{op}} \|\mathbf{x} - \mathbf{z}\| \\ &= \frac{2}{m_i} \|\mathbf{A}_i\|_{\text{op}}^2 \|\mathbf{x} - \mathbf{z}\|. \end{aligned}$$

Thus, $L_i = \frac{2}{m_i} \|\mathbf{A}_i\|_{\text{op}}^2$.

[Non-expansiveness of the proximal step] Let $g(\mathbf{x}) = \lambda \|\mathbf{x}\|_1$. For any $\tau > 0$, the proximal operator $\text{prox}_{\tau g}$ is firmly non-expansive:

$$\|\text{prox}_{\tau g}(\mathbf{u}) - \text{prox}_{\tau g}(\mathbf{v})\| \leq \|\mathbf{u} - \mathbf{v}\| \quad \forall \mathbf{u}, \mathbf{v}.$$

This is a standard property of proximal operators of convex functions.
 [Stability of local LASSO refinement] Under (A2)–(A3), the mapping

$$T_i(\mathbf{x}) = \text{prox}_{\eta\lambda\|\cdot\|_1}(\mathbf{x} - \eta\nabla f_i(\mathbf{x}))$$

is a contraction for each i in a neighborhood of any minimizer of E_i , and the sequence $\mathbf{x}^{(i)}(t+1) = T_i(\mathbf{x}^{(i)}(t))$ is stable in the sense that

$$\|\mathbf{x}^{(i)}(t+1)\| \leq C_i(1 + \|\mathbf{x}^{(i)}(t)\|),$$

for some constant C_i depending on $\|\mathbf{A}_i\|_{\text{op}}$ and $\|\mathbf{y}_i\|$.

[Sketch] Using the Lipschitz continuity of ∇f_i with constant L_i and the standard proximal-gradient analysis, if $0 < \eta < 2/L_i$, then T_i is non-expansive and locally contractive around a minimizer of E_i . The linear growth bound follows from bounding $\|\nabla f_i(\mathbf{x})\|$ in terms of $\|\mathbf{x}\|$ and $\|\mathbf{y}_i\|$, and then using the non-expansiveness of the proximal operator.

A.3 Trust-Based Social Mixing: Consensus and Norm Bounds

Let $\mathbf{X}(t) \in \mathbb{R}^{N \times d}$ be the matrix whose i -th row is $\mathbf{x}^{(i)}(t)$. Define the consensus (mean) state

$$\bar{\mathbf{x}}(t) = \frac{1}{N} \sum_{i=1}^N \mathbf{x}^{(i)}(t),$$

and the centered state matrix

$$\mathbf{E}(t) = \mathbf{X}(t) - \mathbf{1}\bar{\mathbf{x}}(t)^\top.$$

The social mixing update (ignoring local refinement for the moment) is

$$\mathbf{X}(t+1) = \mathbf{S}(t)\mathbf{X}(t).$$

[Preservation of the mean] Under (A4), for any row-stochastic $\mathbf{S}(t)$,

$$\bar{\mathbf{x}}(t+1) = \bar{\mathbf{x}}(t).$$

$$\bar{\mathbf{x}}(t+1) = \frac{1}{N} \sum_{i=1}^N \mathbf{x}^{(i)}(t+1) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N S_{ij}(t) \mathbf{x}^{(j)}(t) = \frac{1}{N} \sum_{j=1}^N \left(\sum_{i=1}^N S_{ij}(t) \right) \mathbf{x}^{(j)}(t).$$

Since $\mathbf{S}(t)$ is row-stochastic, $\sum_i S_{ij}(t) = 1$ for each j , so

$$\bar{\mathbf{x}}(t+1) = \frac{1}{N} \sum_{j=1}^N \mathbf{x}^{(j)}(t) = \bar{\mathbf{x}}(t).$$

[Contraction on the disagreement space] Let $\mathbf{S}(t)$ satisfy (A4) and assume that, when restricted to the subspace orthogonal to $\mathbf{1}$, the operator norm of $\mathbf{S}(t)$ satisfies

$$\|\mathbf{S}(t)\|_{\text{op}, \perp \mathbf{1}} \leq \rho < 1$$

uniformly in t . Then

$$\|\mathbf{E}(t+1)\|_{\text{F}} \leq \rho \|\mathbf{E}(t)\|_{\text{F}},$$

where $\|\cdot\|_F$ is the Frobenius norm. In particular, the disagreement norm decays at least geometrically.

Observe that

$$\mathbf{E}(t+1) = \mathbf{X}(t+1) - \mathbf{1}\bar{\mathbf{x}}(t+1)^\top = \mathbf{S}(t)\mathbf{X}(t) - \mathbf{1}\bar{\mathbf{x}}(t)^\top = \mathbf{S}(t)(\mathbf{X}(t) - \mathbf{1}\bar{\mathbf{x}}(t)^\top),$$

where we used preservation of the mean. Thus,

$$\mathbf{E}(t+1) = \mathbf{S}(t)\mathbf{E}(t).$$

Since each column of $\mathbf{E}(t)$ lies in the subspace orthogonal to $\mathbf{1}$, the operator bound on $\mathbf{S}(t)$ implies

$$\|\mathbf{E}(t+1)\|_F \leq \rho \|\mathbf{E}(t)\|_F.$$

A.4 Combined Local and Social Dynamics

We now consider one outer iteration comprising: (i) local proximal-gradient refinement for each agent, (ii) trust/oracle update producing $\mathbf{S}(t)$, and (iii) social mixing via $\mathbf{S}(t)$.

Let $\mathcal{T}$ denote the combined map

$$\mathbf{X}(t) \mapsto \mathbf{X}(t+1),$$

obtained by applying local maps T_i row-wise, followed by $\mathbf{S}(t)$ mixing. Under (A2)–(A4), for suitably small local step size η and trust matrices satisfying the contraction property above, we obtain a joint contraction in a weighted norm around a neighborhood of $\boldsymbol{\theta}^*\mathbf{1}^\top$. We state a simplified result.

[Local contraction around synchronized Form] Assume (A1)–(A4) and suppose that:

- for each i , T_i is a contraction in a neighborhood of $\boldsymbol{\theta}^*$ with contraction factor $\gamma_i < 1$;
- the trust matrices $\mathbf{S}(t)$ satisfy $\|\mathbf{S}(t)\|_{\text{op}, \perp \mathbf{1}} \leq \rho < 1$ uniformly.

Then there exists a neighborhood $\mathcal{U}$ of the synchronized state $\boldsymbol{\Theta}^* := \mathbf{1}(\boldsymbol{\theta}^*)^\top$ and a constant $\kappa \in (0, 1)$ such that for all $\mathbf{X}(t) \in \mathcal{U}$,

$$\|\mathcal{T}(\mathbf{X}(t)) - \boldsymbol{\Theta}^*\|_F \leq \kappa \|\mathbf{X}(t) - \boldsymbol{\Theta}^*\|_F.$$

[Sketch] We decompose $\mathbf{X}(t)$ into mean and disagreement parts:

$$\mathbf{X}(t) = \mathbf{1}\bar{\mathbf{x}}(t)^\top + \mathbf{E}(t).$$

The local maps T_i act independently and, in a neighborhood of $\boldsymbol{\theta}^*$, contract each row toward $\boldsymbol{\theta}^*$. This yields a contraction on the mean component, up to noise. The trust-based mixing contracts the disagreement component via the previous proposition. Combining these two effects yields a joint contraction in a weighted norm near $\boldsymbol{\Theta}^*$.

A.5 False-Sun Suppression under Oracle Penalties

Finally, we formalize the qualitative behavior that “false suns” (agents aligned with an incorrect subspace) are suppressed by the oracle ensemble.

We model a false-sun agent j as one whose candidate state satisfies $\|\mathbf{x}^{(j)} - \boldsymbol{\theta}^*\| \geq \delta$ for some fixed $\delta > 0$ and for all t in a given window (i.e., it remains far from the true Form). Assume that

the oracle ensemble is aligned with the Form in the sense that there exists $\gamma > 0$ such that for any agent i ,

$$\mathbb{E}[O(\mathbf{x}^{(i)}(t))] \geq \mathbb{E}[O(\boldsymbol{\theta}^*)] + \gamma \|\mathbf{x}^{(i)}(t) - \boldsymbol{\theta}^*\|^2,$$

for a suitable averaged oracle O . Then, for a false-sun agent j , we have

$$\mathbb{E}[O(\mathbf{x}^{(j)}(t))] \geq \mathbb{E}[O(\boldsymbol{\theta}^*)] + \gamma \delta^2.$$

[Expected decay of false-sun trust] Under (A1)–(A5) and the oracle alignment condition above, let $C_{\text{fs}}(t)$ denote the expected trust centrality of a false-sun agent j at time t , i.e.

$$C_{\text{fs}}(t) = \mathbb{E} \left[\sum_{i=1}^N S_{ij}(t) \right].$$

Then, for sufficiently large t and suitable oracle-penalty parameters, there exists $\eta_{\text{fs}} \in (0, 1)$ such that

$$C_{\text{fs}}(t+1) \leq \eta_{\text{fs}} C_{\text{fs}}(t).$$

[Informal sketch] The oracle penalty $\pi_j(t)$ for agent j is a decreasing function of its relative oracle degradation. For a false-sun agent whose state is persistently far from the Form, the oracle cost remains strictly worse in expectation than that of non-adversarial agents. Thus, $\pi_j(t)$ is bounded away from 1 and (on average) decreases relative to other agents. This enters the utility $U_{ij}(t)$ multiplicatively or additively, reducing the softmax mass assigned to j . Over time, as the trust matrices are updated, the relative mass in column j decays geometrically in expectation. The existence of a uniform factor $\eta_{\text{fs}} < 1$ follows from uniform lower bounds on the gap between false-sun and non-false-sun oracle costs together with the parameterization of $\pi_j(t)$ in the softmax.

This formalizes the qualitative phenomenon observed in synthetic experiments: “false suns” may attract temporary trust if they explain local shadows well but are ultimately suppressed by the oracle ensemble as the system converges toward the 25D Form.

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